

Modeling with Systems and Matrices

Answer Key

Algebra 2

Quick Answers

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|-----------------------------------|---|
| 1. $x = 15, y = 8$ | 7. $x = 3, y = 4, x = -4, y = -3$ |
| 2. $x = 8, y = 5$ | 8. (a) $x = 5, y = 70, —$ |
| 3. $x = 12, y = 8$ | 9. $x = 6, y = 9, z = 5$ |
| 4. $x = 2, y = 21, x = 3, y = 24$ | 10. (a) $x = 12, y = 5, x = 5, y = 12, —$ |
| 5. $x = 4, y = 7, z = 5$ | 11. $x = 20, y = 30, z = 50$ |
| 6. (a) $x = 16, y = 8, —$ | 12. (a) $x = 40, y = 40, x = 80, y = 20, —$ |
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What is verified

12 of 16 answers machine-checked against SymPy, across 12 problems.
— marks an answer only you can judge — problems 6, 8, 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-REI.C.6 – *problems 1, 2, 3, 5, 6, 8, 9, 11*

HSA-REI.C.7 – *problems 4, 7, 10, 12*

Difficulty 1–5 of 5 across 16 tagged checks.

1. 3 adult and 2 child tickets cost 61 dollars; 1 adult and 4 child cost 47 dollars. Find the two prices.

Solution: $3x + 2y = 61$ and $x + 4y = 47$. The second equation is already solved for x in one step, so substitute.

$$\begin{aligned}x &= 47 - 4y \\3(47 - 4y) + 2y &= 61 \\141 - 12y + 2y &= 61 \\-10y &= -80\end{aligned}$$

So $y = 8$ and $x = 47 - 32$. Adult
dollars, child
dollars.

$x = 15$
$y = 8$

- 2.** Four crates and three boxes weigh 47 kg; two crates weigh 9 kg less than five boxes. Find each mass.

Solution: $4x + 3y = 47$ and $2x - 5y = -9$. Neither variable is isolated, so eliminate: double the second equation.

$$\begin{aligned} 4x - 10y &= -18 \\ (4x + 3y) - (4x - 10y) &= 47 - (-18) \\ 13y &= 65 \end{aligned}$$

So $y = 5$, and $4x + 15 = 47$ gives $x = 8$. Crate
kg, box
kg.

$x = 8$
$y = 5$

- 3.** 20 kg of blend from coffees at 12 and 18 dollars per kg, costing 288 dollars. Matrix equation and solution.

Solution: The two conditions are $x + y = 20$ (mass) and $12x + 18y = 288$ (cost):

$$\begin{bmatrix} 1 & 1 \\ 12 & 18 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 20 \\ 288 \end{bmatrix}$$

Substituting $y = 20 - x$ into the cost row:

$$\begin{aligned} 12x + 18(20 - x) &= 288 \\ 12x + 360 - 18x &= 288 \\ -6x &= -72 \end{aligned}$$

So
kg of the cheaper coffee and
kg of the dearer one.

$x = 12$
$y = 8$

- 4.** Where does $y = -x^2 + 8x + 9$ meet $y = 3x + 15$?

Solution: Both expressions equal y , so set them equal — one linear and one quadratic equation always reduce to a single quadratic.

$$\begin{aligned} -x^2 + 8x + 9 &= 3x + 15 \\ 0 &= x^2 - 5x + 6 \\ 0 &= (x - 2)(x - 3) \end{aligned}$$

At $x = 2$: $y = 3(2) + 15 = 21$. At $x = 3$: $y = 3(3) + 15 = 24$. The jet meets the wire at

and
(metres).

$x = 2, y = 21$
$x = 3, y = 24$

- 5.** 16 items; cutting 2/3/1 hours totalling 34; finishing 3/1/2 hours totalling 29. Find the counts.

Solution: The three conditions are

$$x + y + z = 16, \quad 2x + 3y + z = 34, \quad 3x + y + 2z = 29$$

Subtracting the first from the second removes z at once:

$$\begin{aligned} (2x + 3y + z) - (x + y + z) &= 34 - 16 \implies x + 2y = 18 \\ (3x + y + 2z) - 2(x + y + z) &= 29 - 32 \implies x - y = -3 \end{aligned}$$

From the second, $x = y - 3$; substituting, $3y = 21$, so $y = 7$, $x = 4$ and $z = 16 - 11$.
Chairs

, tables

, stools

.

$x = 4$

$y = 7$

$z = 5$

6. 24 litres of juice at 3 and 5 dollars per litre costing 88 dollars. (a) Matrix equation and solution. (b) What each row records.

Solution (a):

$$\begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 24 \\ 88 \end{bmatrix}$$

Substituting $y = 24 - x$ into the second row:

$$\begin{aligned} 3x + 5(24 - x) &= 88 \\ -2x &= -32 \end{aligned}$$

So

litres of orange and

litres of mango.

$x = 16$

$y = 8$

Solution (b) — instructor-judged. Row 1 records the volume condition: each litre of either juice counts once toward the 24 litres, which is why both entries are 1. Row 2 records the cost condition, and its entries are the prices per litre. Full credit ties each row to one of the two stated conditions and says what the entries inside a row measure. Describing the columns instead, or restating the two equations, does not answer it.

7. Where does $y = x + 1$ cross $x^2 + y^2 = 25$?

Solution: Substitute the line into the circle — the line is already solved for y , so substitution beats elimination here.

$$\begin{aligned} x^2 + (x + 1)^2 &= 25 \\ 2x^2 + 2x + 1 &= 25 \\ x^2 + x - 12 &= 0 \\ (x - 3)(x + 4) &= 0 \end{aligned}$$

At $x = 3$, $y = 4$; at $x = -4$, $y = -3$. Both points are 5 metres from the centre, as the circle demands. The crossings are

$x = 3, y = 4$

and

$$x = -4, y = -3$$

8. Plan A: 40 dollars + 6 per hour. Plan B: 25 dollars + 9 per hour. (a) Break-even. (b) Which is cheaper at 10 hours.

Solution (a): $y = 6x + 40$ and $y = 9x + 25$. Both are solved for y , so set the right-hand sides equal.

$$\begin{aligned} 6x + 40 &= 9x + 25 \\ 15 &= 3x \end{aligned}$$

So $x = 5$ hours, and $y = 6(5) + 40$. The plans agree at
hours and
dollars.

$$x = 5$$

$$y = 70$$

Solution (b) — instructor-judged. At 10 hours Plan A costs $6(10) + 40 = 100$ dollars and Plan B costs $9(10) + 25 = 115$ dollars, so Plan A is cheaper by 15 dollars. The reason: past the break-even point at 5 hours, the plan with the smaller hourly rate wins, because the extra 15 dollars of fixed charge has by then been repaid by the 3 dollars per hour saved. Full credit needs both costs (or their difference) and the rate-versus-fixed-charge reasoning.

9. 20 drinks at 3, 4, 5 dollars raising 79 dollars, with 3 more mediums than smalls. Find the counts.

Solution: The conditions are $x + y + z = 20$, $3x + 4y + 5z = 79$ and $y = x + 3$. Substituting the third into the other two:

$$\begin{aligned} x + (x + 3) + z &= 20 & \implies & 2x + z = 17 \\ 3x + 4(x + 3) + 5z &= 79 & \implies & 7x + 5z = 67 \end{aligned}$$

From the first, $z = 17 - 2x$; substituting, $7x + 85 - 10x = 67$, so $-3x = -18$. Small
, medium
, large
.

$$x = 6$$

$$y = 9$$

$$z = 5$$

10. A rectangle of perimeter 34 m and area 60 square metres. (a) All pairs. (b) Why they describe one garden.

Solution (a): The conditions are $x + y = 17$ and $xy = 60$. Substituting $y = 17 - x$:

$$\begin{aligned} x(17 - x) &= 60 \\ x^2 - 17x + 60 &= 0 \\ (x - 12)(x - 5) &= 0 \end{aligned}$$

The pairs are
and
(metres).

$$x = 12, y = 5$$

$$x = 5, y = 12$$

Solution (b) — instructor-judged. Swapping x and y changes neither the sum nor the product, so the two conditions cannot tell the two side lengths apart: a 12 by 5 rectangle and a 5 by 12 rectangle are the same garden seen from a different side. Full credit names that symmetry. “They are the same numbers” without saying why the equations cannot distinguish them is only half the answer.

11. 100 kg of three grains; protein 2/1/3 per kg totalling 220; fibre 3/2/1 per kg totalling 170. Solve.

Solution: The blend is

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 100 \\ 220 \\ 170 \end{bmatrix}$$

Eliminate with the mass row:

$$\begin{aligned} (2x + y + 3z) - (x + y + z) &= 220 - 100 & \implies & x + 2z = 120 \\ (3x + 2y + z) - 2(x + y + z) &= 170 - 200 & \implies & x - z = -30 \end{aligned}$$

Subtracting gives $3z = 150$, so $z = 50$, $x = 20$, and $y = 100 - 70$.

kg,
kg,
kg.

$x = 20$
$y = 30$
$z = 50$

12. Three-sided fence of 120 m enclosing 1600 square metres. (a) All pairs. (b) Wider frontage, and are both usable?

Solution (a): The conditions are $x + 2y = 120$ and $xy = 1600$. Substituting $x = 120 - 2y$:

$$\begin{aligned} (120 - 2y)y &= 1600 \\ 2y^2 - 120y + 1600 &= 0 \\ y^2 - 60y + 800 &= 0 \\ (y - 40)(y - 20) &= 0 \end{aligned}$$

At $y = 40$, $x = 40$; at $y = 20$, $x = 80$. The designs are

and

(metres).

$x = 40, y = 40$
$x = 80, y = 20$

Solution (b) — instructor-judged. The second design has the wider river frontage, 80 metres against 40 metres. Both are genuinely usable: $80 + 2(20) = 120$ metres of fence and $80 \times 20 = 1600$ square metres, and likewise for the square design, so neither root is extraneous. Full credit compares the two frontages *and* checks both pairs against both conditions — this is the problem where a student is most tempted to discard a perfectly good second solution.
